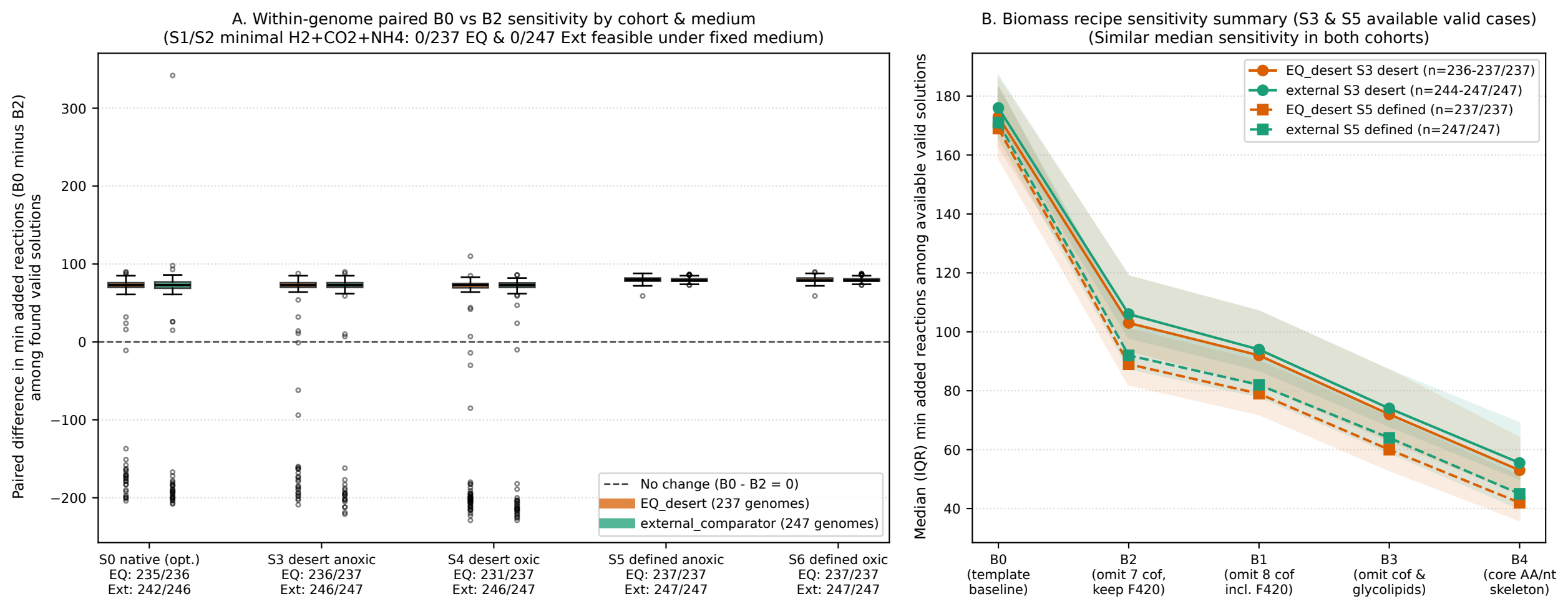
A. Within-genome paired B0 vs B2 sensitivity by cohort & medium
(S1/S2 minimal H2+CO2+NH4: 0/237 EQ & 0/247 Ext feasible under fixed medium)
Paired difference in min added reactions (B0 minus B2)
among found valid solutions
300
200
100
0
−100
−200
No change (B0 - B2 = 0)
EQ_desert (237 genomes)
external_comparator (247 genomes)
S0 native (opt.)
EQ: 235/236
Ext: 242/246
S3 desert anoxic
EQ: 236/237
Ext: 246/247
S4 desert oxic
EQ: 231/237
Ext: 246/247
S5 defined anoxic
EQ: 237/237
Ext: 247/247
S6 defined oxic
EQ: 237/237
Ext: 247/247
B. Biomass recipe sensitivity summary (S3 & S5 available valid cases)
(Similar median sensitivity in both cohorts)
Median (IQR) min added reactions among available valid solutions
180
160
140
120
100
80
60
40
EQ_desert S3 desert (n=236-237/237)
external S3 desert (n=244-247/247)
EQ_desert S5 defined (n=237/237)
external S5 defined (n=247/247)
B0
(template
baseline)
B2
(omit 7 cof,
keep F420)
B1
(omit 8 cof
incl. F420)
B3
(omit cof &
glycolipids)
B4
(core AA/nt
skeleton)